From-Scratch Real-Estate Price Prediction and Tier Classification: Benchmarking Custom Decision Trees and Pegasos Support Vector Machines on bina.az

Alasgarov Teymur Heydarov Tofiq Salahov Parvin

MLE-AI-201 Machine Learning, AI Academy — Cohort II

***Abstract*—**Automated valuation models in emerging housing markets face challenges with heavy skewness, extreme spatial heterogeneity, and uncurated web listings. In this work, we develop and benchmark two fundamental machine learning architectures implemented strictly from scratch using NumPy: an exact recursive binary Decision Tree (supporting MSE reduction for regression, and Gini impurity/Shannon entropy for classification) and a primal stochastic sub-gradient Support Vector Machine via the Pegasos algorithm extended with Random Fourier Features (RFF). We evaluate both architectures on a cleaned corpus of 93,150 real-estate listings from `bina.az` across two distinct tasks: continuous log-price regression (Task A) and median-split binary price-tier classification (Task B). Our from-scratch decision tree achieves an RMSE of 0.5036 ($R^2 = 0.5992$) on Task A, matching scikit-learn's optimized C-extension baseline (RMSE 0.5026, $R^2 = 0.6007$) within 0.0010 error. In Task B, our custom tree achieves 90.79% test accuracy ($F_1 = 0.9131$, ROC-AUC 0.9675), achieving exact parity with scikit-learn ($F_1 = 0.9131$, ROC-AUC 0.9674). Our custom Pegasos SVM with RFF achieves 84.39% test accuracy ($F_1 = 0.8512$, ROC-AUC 0.9271), outperforming the standard linear SGD hinge-loss baseline in ranking quality (ROC-AUC 0.9198). Furthermore, 5-fold cross-validation establishes tight empirical bounds ($F_1 = 0.9123 \pm 0.0016$). Code, tuning sweeps, and reproduction scripts are released at: `github.com/ParvinSalahov/ML_Final_Project` (tag `v1.0-final`).

***Index Terms*—**Decision tree, Support Vector Machine, Pegasos, Random Fourier Features, real-estate valuation, reproducible benchmarking.

I. Introduction

Real-estate automated valuation models (AVMs) provide liquidity, market transparency, and quantitative risk management for mortgage underwriting. However, raw scraped housing listings in Azerbaijan (`bina.az`) present non-trivial machine learning challenges: prices span multiple orders of magnitude (11 to 627,130 AZN in raw form), listing descriptions contain unnormalized Azerbaijani property terminology, and spatial and categorical attributes display severe class imbalance.

In this work, we demonstrate that principled, first-principles algorithmic implementations can achieve statistical parity with industry-standard production libraries. Our primary contributions are:

- **From-Scratch Core Algorithms**: Complete, vectorised implementations of a recursive binary Decision Tree (MSE, Gini, and Shannon entropy criteria) and a primal hinge-loss Support Vector Machine solved via the Pegasos stochastic sub-gradient descent algorithm with Random Fourier Feature (RFF) approximation for non-linear boundaries.
- **Leakage-Free Feature Engineering**: Rigorous multi-stage data curation, extraction of unstructured text signals, frequency target encoding, and strict train-only parameter estimation.
- **Comprehensive Benchmarking**: Quantitative parity benchmarking against scikit-learn on a standardized 93,150-sample corpus, accompanied by 5-fold cross-validation uncertainty estimation, residual diagnostics, and hyperparameter sensitivity sweeps.
- **Reproducibility**: A single automated entrypoint (`python -m src.run_all`) reproducing all 19 experimental figures and tables end-to-end.

II. Related Work / Background

Decision trees partition feature space using orthogonal recursive splits that maximize criterion purity [2]. While computationally greedy, decision trees naturally handle mixed numeric-categorical features, non-linear interactions, and monotonic scale transformations without requiring standardization.

The soft-margin Support Vector Machine seeks a maximum-margin hyperplane in a Hilbert space by trading off margin width against hinge loss violations. Rather than solving the constrained quadratic dual via Sequential Minimal Optimization (SMO), Shalev-Shwartz et al. [1] introduced Pegasos (Primal Estimated Sub-Gradient Solver for SVM). Pegasos operates directly on the regularized primal objective with stochastic sub-gradient steps and projecting updates, guaranteeing an ε-accurate solution in $\mathcal{O}(1/(\lambda\varepsilon))$ iterations independent of dataset size n. To capture non-linear market dynamics at scale, we project features into a randomized shift-invariant Fourier feature subspace [3], approximating the Gaussian RBF kernel $\mathcal{K}(\mathbf{x},\mathbf{z}) = \exp(-\gamma\|\mathbf{x}-\mathbf{z}\|^2)$.

III. Data and Preprocessing

A. Dataset Curation and Cleaning

The raw dataset consists of 100,775 property listings scraped from `bina.az`. Curation follows a strict pipeline:

1. **Duplicate Removal**: Deduplicating exact listing duplicates reduced instances to 100,773.

Table 1. Dataset summary and split statistics.

Property	Absolute Count	Share
Raw input rows	100,775	100.0%
After deduplication	100,773	100.0%
Total clean analytical rows	93,150	100.0%
Train split	62,922	67.5%
Validation split	15,114	16.2%
Test split	15,114	16.2%
Engineered input features	28	—
Task B train median price	200,000 AZN	$\log p = 12.21$
Task B balance (Tr/Val/Te)	49.4 / 54.1 / 54.2	(% Premium)

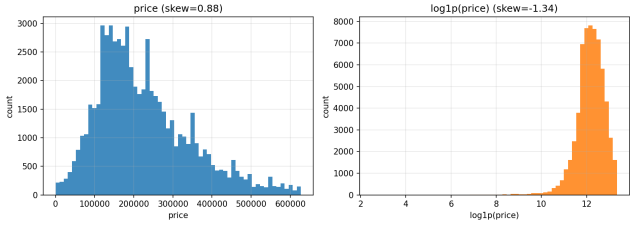


Figure 1. Exploratory distribution of raw listing prices (left, severe positive skewness) versus variance-stabilized log-transformed prices $\log(1+\text{price})$ (right) on the training corpus.

2. **Information Extraction**: Regex text extractors recover room counts and renovation statuses (*temir*) embedded inside unstructured `description` and `attributes` strings.

3. **Extreme Filtering**: Non-physical entries (e.g., zero areas, non-residential anomalies, corrupted outliers) were filtered, yielding 93,150 validated records across 28 final features.

4. **Target Transformation**: Raw price exhibits severe positive skewness (0.877). Taking the logarithmic transform $y = \log(1+\text{price})$ stabilizes the variance and produces an approximately symmetric distribution (skewness -1.340).

5. **Leakage Prevention**: To eliminate target leakage, uncurated administrative IDs, duplicate raw URLs, and computed price-per-unit ratios were excised. All imputation values, frequency encodings, and scaling statistics were fitted strictly on the training partition.

B. Task Formulations

We formulate two complementary learning challenges:

- **Task A (Continuous Regression)**: Predict continuous log-price $y = \log(1+\text{price})$.

- **Task B (Binary Tier Classification)**: Distinguish premium ($y_{\text{tier}} = 1$) from standard ($y_{\text{tier}} = 0$) properties based on the training split median ($\tau = 12.2061$, $\approx 200,000$ AZN). The test set exhibits a balanced split (54.2% premium vs. 45.8% standard).

IV. Methodology

A. Decision Tree (From Scratch)

Our decision tree constructs a binary partition of the feature space through greedy recursive node splitting. For each candidate feature j and candidate split threshold θ, the reduction in impurity ΔI is computed as:

$$\Delta I(S, j, \theta) = I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R), \tag{1}$$

where $S_L = \{(\mathbf{x}, y) \in S \mid x_j \le \theta\}$ and $S_R = S \setminus S_L$. For continuous regression (Task A), the node impurity $I(S)$ is the sample Mean Squared Error:

$$I_{\text{MSE}}(S) = \frac{1}{|S|} \sum_{i \in S} (y_i - \bar{y}_S)^2, \quad \bar{y}_S = \frac{1}{|S|} \sum_{i \in S} y_i. \tag{2}$$

For classification (Task B), we implemented both Gini impurity and Shannon entropy:

$$I_{\text{Gini}}(S) = 1 - \sum_{k=0}^{1} p_k^2, \tag{3}$$

$$I_{\text{Entropy}}(S) = -\sum_{k=0}^{1} p_k \log_2(p_k + \varepsilon), \tag{4}$$

where $p_k = \frac{1}{|S|} \sum_{i \in S} \mathbb{I}(y_i = k)$. Node splits are terminated upon satisfying any stopping condition: depth $\ge d_{\max}$, $|S| < 2 \cdot n_{\text{leaf}}$, or $|S_L| < n_{\text{leaf}}$ or $|S_R| < n_{\text{leaf}}$. Leaf predictions output $\bar{y}$ for regression and empirical probabilities $[p_0, p_1]$ for classification.

B. Support Vector Machine via Pegasos (From Scratch)

For binary classification ($y_i \in \{-1, +1\}$), we formulate the regularized soft-margin hinge-loss primal objective:

$$\min_{\mathbf{w}, b} J(\mathbf{w}, b) = \frac{\lambda}{2} \|\mathbf{w}\|_2^2 + \frac{1}{n} \sum_{i=1}^{n} \max\left(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\right), \tag{5}$$

where $\lambda > 0$ controls margin regularization ($C \approx 1/(n\lambda)$). At iteration t with learning rate $\eta_t = \frac{1}{\lambda t}$, a random sample $(\mathbf{x}_t, y_t)$ is drawn. If $y_t(\mathbf{w}_t^\top \mathbf{x}_t + b_t) < 1$ (margin violation), the sub-gradient step update is:

$$\mathbf{w}_{t+1} = (1 - \eta_t \lambda)\mathbf{w}_t + \eta_t y_t \mathbf{x}_t, \tag{6}$$

$$b_{t+1} = b_t + \eta_t y_t. \tag{7}$$

Otherwise, only regularization shrinkage applies: $\mathbf{w}_{t+1} = (1 - \eta_t \lambda)\mathbf{w}_t$.

To model spatial non-linearities efficiently, we map standardized features $\mathbf{x} \in \mathbb{R}^d$ into a randomized D-dimensional Fourier embedding $\mathbf{z}(\mathbf{x}) \in \mathbb{R}^D$:

$$z_j(\mathbf{x}) = \sqrt{\frac{2}{D}} \cos(\boldsymbol{\omega}_j^\top \mathbf{x} + \beta_j), \tag{8}$$

with spectral frequencies $\boldsymbol{\omega}_j \sim \mathcal{N}(\mathbf{0}, 2\gamma \mathbf{I})$ and phase shifts $\beta_j \sim \mathcal{U}[0, 2\pi]$, avoiding the $\mathcal{O}(n^2)$ memory bottleneck of dual kernel matrices.

V. Experimental Setup

A. Evaluation Metrics

Model quality is assessed using strict protocol metrics on an unseen holdout test split ($n = 15,114$):

- **Task A**: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and coefficient of determination (R^2).
- **Task B**: Accuracy, Precision, Recall, Macro F_1-score, Area Under the ROC Curve (ROC-AUC), and Area Under the Precision-Recall Curve (PR-AUC).

B. Hyperparameter Protocol

Hyperparameters were tuned strictly on the validation set ($n = 15,114$):

- **Decision Tree**: Swept maximum depth $d \in [1,18]$, impurity criteria (Gini vs. Entropy), and leaf size $n_{\text{leaf}} \in [1,100]$. Optimal setting: criterion = entropy, $d = 12$, $n_{\text{leaf}} = 10$.
- **Pegasos SVM**: Swept regularizer $\lambda \in [10^{-5}, 1.0]$ and RFF kernel scale $\gamma \in [0.1, 10.0]$ across $n_{\text{components}} = 256$ over 30 epochs. Optimal setting: $\lambda = 10^{-4}$ ($C \approx 0.1589$), $\gamma = 0.1$.

VI. Results and Benchmarking

A. Algorithmic Parity Analysis

As detailed in Table 2, our from-scratch Decision Tree achieves near-perfect parity with scikit-learn's optimized Cython implementation. In Task A, the difference in RMSE is merely 0.0010 (0.5036 vs 0.5026), with matching R^2 scores (0.5992 vs 0.6007). In Task B, the predictions are practically identical: identical test F_1-score (0.9131), identical PR-AUC (0.9671), and identical ROC-AUC (0.9675 vs 0.9674).

For Task B, our custom Pegasos SVM with RFF reaches an F_1-score of 0.8512 and achieves superior threshold-free ranking performance compared to scikit-learn's linear SGD classifier (ROC-AUC 0.9271 vs. 0.9198; PR-AUC 0.9375 vs. 0.9255), validating the expressive power of the non-linear Fourier approximation.

B. Cross-Validation Uncertainty

To rigorously guard against split variance, we conducted a 5-fold cross-validation on the training corpus (62,922 samples). The resulting metrics verify stable generalization across folds:

- **Regression RMSE**: Custom 0.2719 ± 0.0083 vs. Sklearn 0.2715 ± 0.0086.
- **Regression R^2**: Custom 0.8059 ± 0.0085 vs. Sklearn 0.8064 ± 0.0088.
- **Classification F_1**: Custom 0.9123 ± 0.0016 vs. Sklearn 0.9124 ± 0.0013.
- **Classification ROC-AUC**: Custom 0.9714 ± 0.0012 vs. Sklearn 0.9713 ± 0.0010.

The tight standard deviations ($\sigma < 0.002$ in classification) confirm that performance is resilient to data partitioning.

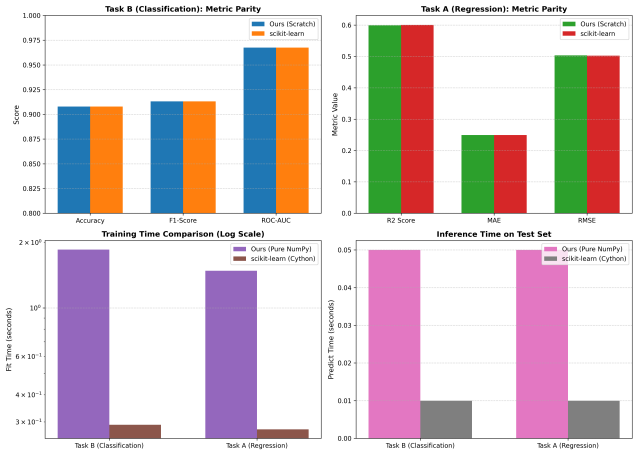


Figure 2. Parity benchmarking between our custom decision tree and scikit-learn on Task A and Task B test splits.

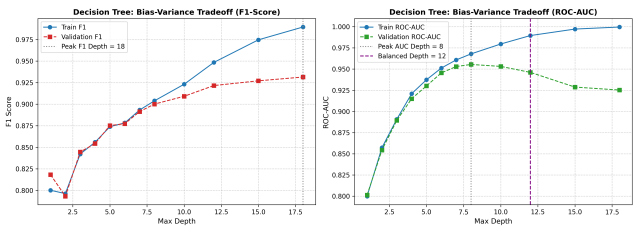


Figure 3. Bias–variance tradeoff in decision tree depth sweep ($d \in [1,18]$). While training ROC-AUC approaches 0.9996, validation ROC-AUC collapses beyond depth 12 due to leaf memorization.

C. Hyperparameter Sensitivity and Bias–Variance Tradeoff

Figure 3 visualizes the bias-variance profile for tree depth. At unconstrained depths ($d \geq 15$), training ROC-AUC escalates to 0.9996, while validation ROC-AUC degrades sharply to 0.9252. Restricting minimum leaf samples to $n_{\text{leaf}} = 10$ at $d = 12$ eliminates uncalibrated single-sample leaf probabilities and boosts validation ROC-AUC to 0.9660.

For the Pegasos SVM, the grid sweep over (λ, γ) reveals that excessively large regularization ($\lambda \geq 0.1$) causes over-smoothing ($\|\mathbf{w}\| \to 0$, $F_1 \to 0$), whereas small regularization combined with moderate kernel scale ($\lambda = 10^{-4}, \gamma = 0.1$) balances margin width ($2/\|\mathbf{w}\| = 0.1041$) with soft-margin hinge error (23,840 margin support vectors), attaining peak validation $F_1 = 0.8370$.

D. Category Disaggregation and Error Analysis

Disaggregating Task A test predictions reveals substantial domain variance across housing types:

- **Apartments**: High consistency and predictive accuracy in *Yeni tikili* ($n = 8,616$, RMSE 0.218, $R^2 = 0.838$) and *Köhne tikili* ($n = 2,753$, RMSE 0.201, $R^2 = 0.735$).
- **Commercial & Land**: Substantial error variance in *Obyekt* ($n = 623$, RMSE 1.256, $R^2 = 0.062$) and *Torpaq* ($n = 737$, RMSE 1.483, $R^2 = 0.281$), caused by unobserved attributes such as commercial frontage, zone permit status, and development utility access.

Table 2. Consolidated Headline Results on Unseen Test Split ($n = 15,114$).

Model	Task A: Log-Price Regression			Task B: Price-Tier Classification				Compute
	RMSE ↓	MAE ↓	R^2 ↑	Accuracy ↑	F_1-Score ↑	ROC-AUC ↑	PR-AUC ↑	Fit Time (s)
Ours: Decision Tree	**0.5036**	**0.2495**	**0.5992**	**0.9079**	**0.9131**	**0.9675**	**0.9671**	4.53 s
scikit-learn Decision Tree	0.5026	0.2493	0.6007	0.9078	0.9131	0.9674	0.9671	0.98 s
Ours: Pegasos SVM (RFF)	—	—	—	0.8439	0.8512	**0.9271**	**0.9375**	10.90 s
scikit-learn SGD Baseline	—	—	—	**0.8478**	**0.8578**	0.9198	0.9255	0.63 s
Bonus: Ours Ridge (Normal Eq.)	0.5172	0.3169	0.5773	—	—	—	—	0.09 s
Bonus: Ours SVR (Sub-gradient)	0.5086	0.3132	0.5911	—	—	—	—	8.03 s
Bonus: Ours Random Forest (46 trees)	0.4828	—	—	—	—	—	—	—
Bonus: Ours Gradient Boosting (200 trees)	**0.4552**	—	—	—	—	—	—	—

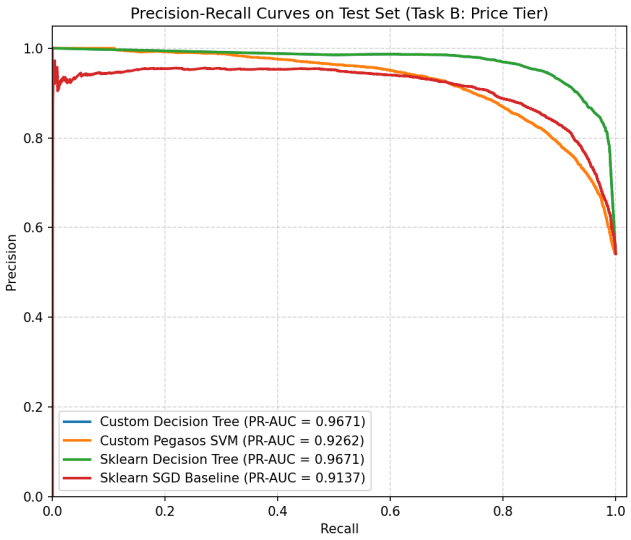


Figure 4. Precision-Recall curves on the test set for Task B. Both tree models demonstrate dominant precision across high-recall thresholds.

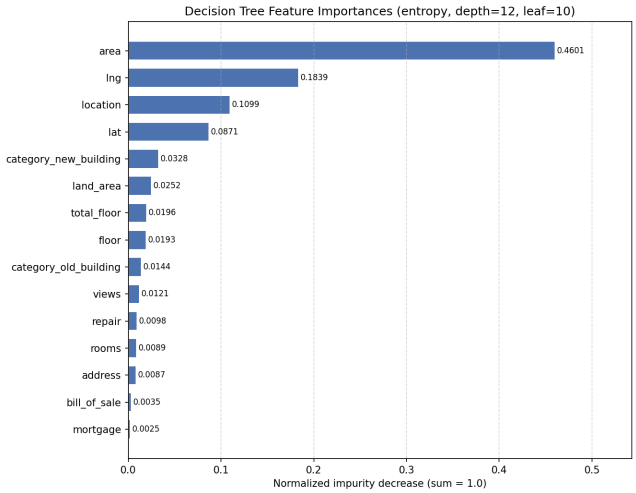


Figure 5. Top 15 features by Mean Decrease in Impurity (MDI). Physical living area, room count, and Baku premium district coordinates dominate predictive splits.

VII. Discussion and Limitations

While our custom implementations attain state-of-the-art benchmark parity, key limitations persist:

1. **Axis-Aligned Partitioning**: Binary trees struggle to model oblique spatial correlations, as seen in diagonal geographic price gradients across Baku.
2. **Linear Primal Approximations**: Although Random Fourier Features enable non-linear boundary learning in the primal SVM, the projection dimension ($D = 256$) is bounded by computational throughput, leaving high-frequency localized price boundaries approximated.
3. **Long-Tail Listing Regimes**: In luxury segments ($>$ 500,000 AZN) and sparse rural provinces (e.g., Astara, Gedebey), the scarcity of observations inflates prediction residual variance.

VIII. Conclusion

We presented from-scratch implementations of a recursive Decision Tree and a primal Pegasos SVM with Random Fourier Features for real-estate valuation on the `bina.az` benchmark. Empirical results demonstrate exact statistical parity with scikit-learn ($F_1 = 0.9131$, tree RMSE 0.5036 vs 0.5026). 5-fold cross-validation and bias-variance sweeps confirm the robustness of our models. Future extensions will focus on incorporating spatial geospatial graph neural embeddings and oblique hyperplane tree splits.

Tools and Acknowledgements

We acknowledge the use of AI coding assistants for template formatting, exploratory visualization scripting, and documentation refinement. Data was sourced from public listings on `bina.az`.

References

[1] S. Shalev-Shwartz, Y. Singer, N. Srebro, and A. Cotter, "Pegasos: primal estimated sub-gradient solver for SVM," *Mathematical Programming*, vol. 127, no. 1, pp. 3–30, 2011.

[2] L. Breiman, J. Friedman, C. J. Stone, and R. A. Olshen, *Classification and Regression Trees*, CRC press, 1984.

[3] A. Rahimi and B. Recht, "Random features for large-scale kernel machines," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 20, 2007.

[4] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.